Elements of AIML
Unit 02 – Lecture 01 Notes
Propositional Logic Foundations (Entailment)

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1 Lecture Overview

These notes introduce **propositional logic** as a tool to represent knowledge and to reason correctly from that knowledge. In AI, we often store:

- **facts** (what is currently true), and
- **rules** (how truths imply other truths),

and then ask queries such as: "Given my knowledge base, must this query be true?".

The key concept for that question is **entailment**:

$$KB \models \alpha \quad \text{(every model of } KB \text{ is a model of } \alpha\text{)}.$$

This lecture also introduces **CNF** (conjunctive normal form), because many automated reasoning methods expect knowledge in a standard clause-like form (resolution is covered later in this unit).

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.

2 How to use these notes (self-study)

Read the sections in order. When you see a worked example, try to solve it yourself before reading the solution. For practice problems, write your full reasoning (not only the final answer); in logic, the *justification* is part of the learning outcome.

3 Learning Outcomes

- Define propositions and logical connectives; compute truth values using truth tables.
- Classify formulas as valid (tautology), satisfiable, or unsatisfiable.
- Explain semantic entailment $KB \models \alpha$ and test it for small KBs using models/truth tables.
- Convert simple propositional formulas to CNF and explain why CNF is useful.
- Explain (at a high level) why propositional logic is insufficient for many-object relational domains.

4 Keywords

- **Proposition:** a declarative statement that is either True or False.

- **WFF (well-formed formula):** a syntactically valid logic expression.
- **Model:** a truth assignment that makes a sentence true.
- **Entailment ($\models$):** truth in all models (semantic consequence).
- **Validity:** true in all models (tautology).
- **Satisfiable:** true in at least one model.
- **CNF:** conjunction (AND) of disjunctions (OR-clauses).

5 Abbreviations and symbols

- KB: knowledge base (set of sentences/facts/rules).
- α: query/goal sentence.
- $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$: NOT, AND, OR, IMPLIES, IFF.
- $\models$: entailment (semantic); $\vdash$: derivation/provability (syntactic).
- **CNF**: conjunctive normal form; **literal**: p or $\neg p$; **clause**: OR of literals.

6 Why logic in AI? (KBs and queries)

6.1 Knowledge base and query

A **knowledge base (KB)** is a set of sentences describing a world. For example:

- Fact: $p =$ "It is raining."
- Rule: $p \rightarrow q =$ "If it is raining, then the ground is wet."

A **query** α is what we want to know, e.g., $q =$ "The ground is wet.".

6.2 Syntax vs semantics (the important separation)

- **Syntax**: which strings count as legal formulas (grammar).
- **Semantics**: what those formulas mean (truth in models).

We can manipulate syntax mechanically (proofs) and still preserve meaning if our inference rules are sound.

7 Propositional logic

7.1 Syntax: propositions and well-formed formulas

Propositional logic uses propositional symbols (atoms) like p, q, r. We build complex formulas using connectives:

$$\neg p,\ (p \wedge q),\ (p \vee q),\ (p \rightarrow q),\ (p \leftrightarrow q)$$

A formula is a **well-formed formula (WFF)** if it follows the grammar (symbols, connectives, parentheses).

7.2 Semantics: truth assignments and models

A **truth assignment** v maps each propositional symbol to T or F. The truth of a complex formula is computed using the connective truth tables.

If a formula φ is true under assignment v, we say v is a **model** of φ and write $v \models \varphi$. If v makes all sentences in KB true, then v is a model of KB (treating KB as a conjunction of its sentences).

7.3 Truth tables and classification of formulas

Truth tables list all possible assignments and compute the formula value. With n propositional symbols, there are 2^n rows.

Using a truth table we can classify:

- **Tautology/valid**: always true (e.g., $p \vee \neg p$).
- **Contradiction/unsatisfiable**: always false (e.g., $p \wedge \neg p$).
- **Satisfiable**: true in at least one assignment (has a model).
- **Unsatisfiable KB**: no assignment satisfies all KB sentences (KB is inconsistent).

7.4 Implication ($p \to q$) and common confusions

In everyday language, "if" sometimes implies causation. In propositional logic, implication is purely truth-functional:

$$(p \to q) \text{ is false only when } p \text{ is true and } q \text{ is false.}$$

The most useful equivalence:

$$(p \to q) \equiv (\neg p \vee q)$$

This equivalence is used in CNF conversion and SAT/resolution methods.

8 Entailment ($KB \models \alpha$)

8.1 Definition (semantic)

KB entails α (written $KB \models \alpha$) iff:

> In *every* model where KB is true, α is also true.

Equivalently: there is **no counterexample model** where KB is true and α is false.

8.2 How to test entailment in propositional logic

For propositional logic, the following are equivalent (and can be checked by truth tables for small KBs):

- $KB \models \alpha$
- $(KB \to \alpha)$ is valid (a tautology), where KB is treated as a conjunction of its sentences
- $KB \wedge \neg\alpha$ is **unsatisfiable**

The last form is important because many automated reasoning methods reduce inference to satisfiability.

8.3 Entailment vs implication vs derivation (do not mix them up)

- **Implication** ($\rightarrow$) is a connective inside the language.
- **Entailment** ($\models$) is a semantic relationship (about all models).
- **Derivation** ($\vdash$) is a syntactic relationship (there exists a proof using rules).

Soundness and completeness (intuition). A proof system is **sound** if it never proves a sentence that is not entailed. It is **complete** if it can prove every sentence that is entailed.

8.4 Worked example: model-based reasoning

Let:

$$KB = \{(p \rightarrow q),\ (q \rightarrow r),\ p\}, \qquad \alpha = r$$

Reasoning using rules:

1. From $(p \rightarrow q)$ and p, derive q (modus ponens).
2. From $(q \rightarrow r)$ and q, derive r.

So every model of KB must make r true, hence $KB \models r$.

If you wanted to check via models (truth table), you would enumerate assignments of (p, q, r), keep those that satisfy all three KB sentences, and verify that in all such rows, $r = T$.

9 Normal forms and CNF (why we care)

9.1 Literals, clauses, and CNF

- A **literal** is p or $\neg p$.
- A **clause** is an OR of literals, e.g., $(\neg p \vee q \vee r)$.
- **CNF** is an AND of clauses, e.g.,

$$(\neg p \vee q) \wedge (p \vee r) \wedge (\neg r)$$

9.2 CNF conversion recipe (propositional)

To convert a formula to CNF:

1. Eliminate $\leftrightarrow$ and $\rightarrow$.
2. Push negations inward (De Morgan + double negation).
3. Distribute $\vee$ over $\wedge$ until you get an AND of ORs.
4. Flatten nested AND/OR groups.

9.3 Worked CNF example (distribution is required)

Convert $p \rightarrow (q \wedge r)$:

1. Eliminate implication:

$$\neg p \vee (q \wedge r)$$

2. Distribute $\vee$ over $\wedge$:

$$(\neg p \vee q) \wedge (\neg p \vee r)$$

This is now in CNF (two clauses).

10 Limits of propositional logic (preview of predicate logic)

Propositional logic is not designed for "many objects + relations" worlds. It lacks:

- **relational structure** (e.g., $Parent(x, y)$),
- **quantifiers** ("all" / "some"),
- **compact general rules** that apply to arbitrarily many objects.

Predicate (first-order) logic addresses this by adding objects, predicates, and quantifiers. We cover that fully in **Unit 02 Lecture 02**.

11 Common pitfalls (and how to avoid them)

- **Mixing up $\models$ and $\rightarrow$:** $KB \models \alpha$ is a semantic relationship (about all models); $KB \rightarrow \alpha$ is a formula inside the language.
- **Forgetting the meaning of "model of a KB":** a model of KB is a truth assignment that makes *every* sentence in KB true (treat KB like a conjunction).
- **Treating "satisfiable" as "true":** satisfiable means "true in at least one assignment"; it does not mean "true in the real world".
- **CNF conversion mistakes:** eliminate $\leftrightarrow$ and $\rightarrow$ first, then push negations inward, *then* distribute $\vee$ over $\wedge$.
- **Implication intuition trap:** $p \rightarrow q$ does not mean causation; it is false only when p is true and q is false.

12 Practice (with answers)

P1. Build a truth table and decide if $(p \rightarrow q) \leftrightarrow (\neg p \vee q)$ is valid.

Answer: It is valid. Both expressions have identical truth values for all assignments of (p, q).

P2. Give a satisfying assignment for $(p \vee q) \wedge (\neg p \vee r) \wedge (\neg r)$.

Answer: $r = F$ (from $\neg r$), then $\neg p$ must be true so $p = F$, then $q = T$ to satisfy $(p \vee q)$. One model: $(p = F, q = T, r = F)$.

P3. Show that $KB = \{(p \rightarrow q), (q \rightarrow r), p\}$ entails r.

Answer: Modus ponens yields q, then yields r, so $KB \models r$.

P4. Convert to CNF: $p \leftrightarrow q$.

Answer: $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv (\neg p \vee q) \wedge (\neg q \vee p)$.

P5. Convert to CNF: $p \to (q \wedge r)$.

Answer: $(\neg p \vee q) \wedge (\neg p \vee r)$.

P6. Give one limitation of propositional logic in real knowledge bases.

Answer: it cannot express general rules like "all students..." without writing separate symbols/rules for each object.

13 Exit questions (with answers)

Q1. What does $KB \models \alpha$ mean (in one sentence)?

Answer: In every model where all sentences of KB are true, α is also true.

Q2. For a small KB, how can we test entailment using models/truth tables?

Answer: Enumerate truth assignments; keep those that satisfy KB; check that α is true in all of them (no counterexample).

Q3. What is one common confusion about $p \to q$?

Answer: Confusing it with causation; logically it is false only when p is true and q is false.

Q4. Give one limitation of propositional logic that motivates predicate logic.

Answer: Propositional logic cannot compactly express relations and quantifiers (many-object rules).

14 Summary

Propositional logic provides a precise language for facts and rules, and **entailment** defines what it means for a query to follow from a knowledge base. Truth tables provide a brute-force entailment test for small problems. CNF is a standard representation that prepares formulas for automated inference procedures.

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.

15 Further Reading

- Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), chapters on logical agents.
- Huth and Ryan, *Logic in Computer Science* (2e), propositional logic chapters.

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.

16 Next Lecture Preview

Next (Unit 02 Lecture 02): Predicate (first-order) logic — predicates, quantifiers, translating English to logic, and scope pitfalls.